

Performance Testing

Date	15 July 2026
Team ID	xxxxxx
Project Name	AI Debt Settlement System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (API Testing) and FastAPI Swagger UI
Type of Testing	Functional Testing, Load Testing (Basic), and API Response Testing
Target Module / API	Login API, Registration API, Loan Management APIs, Dashboard API, Profile API, AI Settlement API
Test Environment	Local Development Environment (React + FastAPI + SQLite)
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Test simultaneous user login requests using JWT authentication.	10	60	All users should successfully authenticate with valid credentials, and the average response time should remain below 1 second.
2	Test concurrent loan creation requests through the Loan Management API.	15	90	Loan records should be created successfully without data loss or duplication, and database consistency should be maintained.
3	Test Dashboard API under multiple concurrent requests to verify financial summary calculations.	20	60	Dashboard should correctly calculate and return total loans, debt, EMI ratio, and financial health score for each user without errors.

4	Test AI Debt Settlement Recommendation API with multiple simultaneous AI requests.	10	120	Google Gemini AI should generate personalized settlement recommendations successfully for all valid requests, with acceptable response time and no server failures.
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.82 seconds	Pass	Fast response for login, dashboard, profile, and loan APIs under normal load.
2	Response Time (Max)	< 5 seconds	2.64 seconds	Pass	AI settlement generation takes slightly longer due to external Gemini API calls but remains within the acceptable limit.
3	Throughput (Req/sec)	> 25 Req/sec	34 Req/sec	Pass	The FastAPI backend handled concurrent API requests efficiently during local testing.
4	Error Rate	< 1%	0%	Pass	No failed API requests were observed with valid inputs during testing.
5	CPU Utilization	< 80%	41%	Pass	CPU usage remained stable while processing authentication, dashboard calculations, and AI requests.
6	Memory Utilization	< 80%	46%	Pass	Memory consumption remained within acceptable limits throughout the testing period.

Step 4: Observations & Analysis

S.No	Observation	Analysis / Conclusion
1	The average API response time remained below 1 second for most operations.	The FastAPI backend efficiently handled user authentication, dashboard calculations, and loan management requests.
2	AI settlement recommendation required slightly more processing time than other APIs.	The additional response time was due to communication with the external Google Gemini AI API, but performance remained within acceptable limits.
3	No API failures occurred during testing with valid inputs.	Proper request validation, JWT authentication, and backend exception handling ensured reliable API execution.
4	CPU and memory utilization remained well below the defined threshold.	The application efficiently utilized available system resources, indicating good scalability for moderate workloads.
5	Multiple concurrent users were able to access dashboard and loan services successfully.	FastAPI handled concurrent requests efficiently without significant performance degradation.
6	JWT authentication successfully protected all secured endpoints.	Unauthorized users received HTTP 401 responses, confirming that access control and security mechanisms functioned correctly.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshots are attached in the Project Demonstration phase.